

Option Trading Strategies

The choices a trader makes depends on the trader's judgement about how prices will move and the trader's willingness to take risks.

What are the three most basic investment objectives?

1. safety (hedging/ risk management),
2. income (speculation),
3. growth (income generation)

Being **bearish** in trading means you believe that a market, asset or financial instrument is going to experience a **downward** trajectory. Being bearish is the opposite of being **bullish**, which means that you think the market is heading **upwards**.

Why are option strategies useful?

If you only trade equities, your primary trading strategies are limited to buying the stock if you are bullish or shorting the stock if you are bearish.

By using option strategies, you'll get more ways to trade whether you're bullish, bearish, or even neutral on the market.

The benefits of using option strategies include:

- Lock in the price of a stock for a specified time period without having to commit to buying the stock.
- Buy a stock in the future at a discount to its current price.
- Protect a stock in your portfolio from a substantial price decline.
- Generate income on existing stocks in your portfolio.

Trading an option and the underlying asset

For convenience, we will assume that the asset underlying the options considered is a stock.

Profit from a trading strategy is the final payoff minus the initial cost of establishing the strategy without any discounting.

Protective Put

Investor would need to **long a stock** and simultaneously **hold a put** on the same stock.

The objective of a protective put strategy is to provide downside protection for an existing stock position. This strategy is primarily used to safeguard unrealized gains and limit potential losses in a declining market.

The protective put strategy is used when an investor holds a moderately bullish outlook on a stock but wants to protect against adverse market movements. This strategy is suitable when the investor is bullish about the market but cautious about potential downside risk.

The investor starts by owning shares of a particular stock in their portfolio. These shares are often referred to as the "underlying asset."

To protect against potential declines in the stock's value, the investor simultaneously purchases put options on the same stock. These put options give the investor the right (but not the obligation) to sell the stock at a predetermined strike price.

The strike price of the put is a crucial consideration. It is typically chosen at a level that represents the maximum acceptable loss for the investor. This means that if the stock's price falls below the strike price of the put, the investor can sell the stock at the higher strike price, thereby limiting losses.

The investor pays a premium for purchasing the put options, which is the cost of obtaining downside protection.

Scenario 1: Bullish Outcome: If the stock price rises or remains stable, the investor can continue to hold the stock and let the put options expire worthless. While they incur the cost of the put premium, their overall gains from the stock position are not limited. Total payoff would be ST and the profit would be $[ST - (So + Po)]$.

Scenario 2: Bearish Outcome: In the event of a decline in the stock's price, the protective put comes into play. The investor can sell the stock at the strike price of the put, even if the market price is lower. Total payoff would be X and the profit would be $[X - (So + Po)]$.

The protective put strategy is a risk management tool that allows investors to maintain a bullish outlook on a stock while protecting against potential losses. It provides a clear downside limit, making it a popular choice for investors seeking to protect their investments in uncertain markets.

Covered Call

Investor would need to **long a stock** and **write a call** on the same stock.

This is a written call option when the shares that might have to be delivered are already owned.

Covered calls are far less risky than writing calls naked because the worst that can happen is that the investor is required to sell shares already owned at below their market value.

A covered call is an options trading strategy where an investor holds a long position in an underlying asset, such as a stock, and simultaneously writes a call option on that same asset.

The objective of a covered call strategy is to generate income from the premiums received by selling the call option while still owning the underlying asset. It also protects the writer of the call if the option is-in-the-money and they are required to “deliver” the stock to the call option holder.

The outlook is generally neutral to bearish.

The payoff for a covered call strategy is characterized by limited upside potential and reduced downside risk compared to owning the stock alone. If the stock price remains below the strike price at expiration, the investor keeps the premium received from writing the call. However, if the stock price rises significantly above the strike price, the profit potential is capped because the investor is obligated to sell the stock at the call's strike price.

Scenario 1: Bullish Outcome: If the stock price rises, the call will be exercised, and the investor will be required to “sell” the stock to the option holder at the strike price. Total payoff would be X and total profit would be $[X - (S_0 - C_0)]$.

Scenario 2: Bearish Outcome: In the event of a decline in the stock's price, the call option would lapse and expire worthless. Total Payoff would be S_T and total profit would be $[S_T - (S_0 - C_0)]$.

Collar

A collar strategy involves both a covered call and a protective put.

An investor would therefore need to **long the stock, hold a put** – with a strike price below the current stock price (this provides downside protection), and **write a call** – with a strike price above the current stock price (generates income through receipt of the premium).

Collars are a risk management tool. Investors may use them to limit the downside risk associated with owning a particular stock. By holding a put option, they have the right to sell the stock at a predetermined strike price, thus establishing a floor for potential losses.

Writing the call option in a collar generates income in the form of the premium received. Investors can use this income to offset the cost of purchasing the protective put or as additional profit. It essentially allows them to get paid while protecting their position.

The ideal market outlook for a collar strategy is typically cautious or neutral. You want to protect your gains but don't necessarily expect significant price movements in either direction. It's not ideal for strongly bullish scenarios as it limits potential upside due to the call option obligation.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises above X_2), you benefit from the stock's gains up to the call option's strike price. Beyond that strike price, your profit potential is limited because you are obligated to "sell" the stock at the call options' strike price. Total payoff would be X_2 and profit would be $[X_2 - (S_0 + P_0 - C_0)]$.

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls below X_1), your potential loss is limited because you have the right to sell the stock at the put option's strike price. This provides a floor to your losses. Total payoff would be X_1 and profit would be $[X_1 - (S_0 + P_0 - C_0)]$.

Scenario 3: (S_T lies in between X_1 and X_2). Both the call and the put would be out-of-the-money and would not be exercised. Total payoff would be S_T and profit would be $[S_T - (S_0 + P_0 - C_0)]$.

Combinations

A combination is an option trading strategy that involves taking a position in both calls and puts on the same stock (underlying asset)

Straddle

Involves simultaneously holding a **European call** and **holding a European put** with the same strike price and the same expiration date for the same underlying asset.

A straddle is appropriate when an investor is expecting a large move in a stock price but does not know in which direction the move will be. The ideal market outlook for a long straddle is one of high volatility or uncertainty. You're looking for a significant price change, regardless of whether it's an upward or downward movement. It's not ideal for markets with low volatility, as the cost of buying both a call and a put option can be relatively high.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises significantly), you can benefit from the call option as it allows you to buy the stock at the lower strike price. However, the put option will expire worthless, and the cost of buying both options will be a factor in your overall profit. Total payoff would be $(S_T - X)$ and profit: $[(S_T - X) - (C_o + P_o)]$.

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls significantly), you can benefit from exercising the put option as it allows you to sell the stock at the higher strike price. The call option will expire worthless. Total payoff would be $(X - S_T)$ and profit: $[(X - S_T) - (C_o + P_o)]$.

Strangle

A long **strangle** is an options trading strategy where an investor simultaneously **holds a call option** with a higher strike price and **holds a put option** with a lower strike price, both with the same expiration date on the same underlying asset.

The primary objective of a long strangle is to profit from significant price movement in the underlying asset. It's used when you expect a substantial price swing but are unsure about the direction (up or down).

The ideal market outlook for a long strangle is one of high volatility or uncertainty. You're looking for a significant price change, and it doesn't matter much whether it's an upward or downward movement. It's not suitable for markets with low volatility, as the combined cost of buying both a call and a put option can be relatively high.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises significantly above X_2), you can exercise the call option as it allows you to buy the stock at X_2 . The put option expires worthless. Total payoff is $(S_T - X_2)$ and profit would be $[(S_T - X_2) - (P_o + C_o)]$, where P_o is the premium for the put with a strike price at X_1 and C_o is the premium for the call with a strike price at X_2 .

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls significantly below X_1), you can exercise the put option as it allows you to sell the stock at X_1 . The call option expires worthless. Total payoff is $(X_1 - S_T)$ and profit would be $[(X_1 - S_T) - (P_o + C_o)]$, where P_o is the premium for the put with a strike price at X_1 and C_o is the premium for the call with a strike price at X_2 .

Scenario 3: (S_T lies in between X_1 and X_2): Both the call and the put option would expire worthless. Payoff would be zero and loss would be $0 - (P_o + C_o)$, where P_o is the premium for the put with a strike price at X_1 and C_o is the premium for the call with a strike price at X_2 .

Strips and Straps

A **strip** consists of a long position in **one European call and two European puts** with the same strike price and expiration date.

In a **strip** the investor is betting that there will be a big stock price move and considers a **decrease** in the stock price to be **more likely** than an increase.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises significantly), you can exercise the call option as it allows you to buy the stock at X . The two put option expire worthless. Total payoff is $(S_T - X)$ and profit would be $[(S_T - X) - (2P_o + C_o)]$.

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls significantly), you can exercise both put options as it allows you to sell stocks at X . The call option expires worthless. Total payoff is $2(X - S_T)$ and the profit would be $[(2X - 2S_T) - (2P_o + C_o)]$.

A **strap** consists of a long position in **two European calls and one European put** with the same strike price and expiration date.

In a **strap** the investor is also betting that there will be a big stock price move. However, in this case, an **increase** in the stock price is considered to be **more likely** than a decrease.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises significantly), you can exercise both call options as it allows you to buy stocks at X. The put option expires worthless. Total payoff is $2(ST-X)$ and profit would be $[2(ST-X)-(Po+2Co)]$.

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls significantly), you can exercise the put option as it allows you to sell the stock at X. The call options expire worthless. Total payoff is $(X-ST)$ and the profit would be $[(X-ST) - (Po+2Co)]$.

Spreads

Spreads involve either taking a position in two or more calls or taking a position in two or more puts.

Bull Spread

A **bull spread** can be created by **holding a call with a lower** strike price and **writing a call with a higher** strike price with the same expiration date and on the same underlying asset.

The objective of a bull spread is to benefit from a moderately bullish market outlook. It's used when you expect the price of the underlying asset to increase but not dramatically. This strategy allows you to capitalize on a potential price rise while limiting your upfront costs and potential losses.

The ideal market outlook for a bull spread is one of moderate optimism. You anticipate a price increase in the underlying asset, but you don't expect it to rise significantly or rapidly. It's not suited for strongly bearish or highly volatile market scenarios.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises significantly above X_2), you can exercise both the call option held at X_1 and written at X_2 . The total payoff is (X_2-X_1) and the profit

would be $(X_2 - X_1) + C_2 - C_1$, where C_2 is the premium for the call written, and C_1 is the premium for the call held.

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls significantly below X_1). Neither of the call options would be exercised, they would both lapse and expire worthless. Total payoff would be zero and profit would be: $0 + C_2 - C_1$, where C_2 is the premium for the call written, and C_1 is the premium for the call held.

Scenario 3: (ST lies in between X_1 and X_2): The call option held with a strike price of X_1 will be exercised while the call option written with a strike price of X_2 would not be exercised and would lapse and expire worthless. Total payoff is $(S_T - X_1)$ and profit would be $(S_T - X_1) + C_2 - C_1$, where C_2 is the premium for the call written, and C_1 is the premium for the call held.

Bear Spread

A **bear spread** is an options trading strategy that involves **holding a put option with a higher** strike price and simultaneously **writing a put option with a lower** strike price, both with the same expiration date and on the same underlying asset.

The objective of a bear spread is to benefit from a moderately bearish market outlook. It's used when you expect the price of the underlying asset to decrease but not significantly. This strategy allows you to capitalize on a potential price decline while limiting your upfront costs and potential losses.

The ideal market outlook for a bear spread is one of moderate pessimism. You anticipate a price decrease in the underlying asset, but you don't expect it to fall significantly or rapidly. It's not suited for strongly bullish or highly volatile market scenarios.

Scenario 1: Bullish Outcome: In a bullish scenario (the stock price rises significantly above X_2), both out options would lapse and expire worthless and not be exercised. Total payoff would be zero and profit would be: $0 + P_1 - P_2$, where P_1 is the premium for the put option written and P_2 is the premium for the put option held.

Scenario 2: Bearish Outcome: In a bearish scenario (the stock price falls significantly below X_1). Both put options would be exercised. Total payoff would be $(X_2 - X_1)$ and profit would be: $[(X_2 - X_1) - P_2 + P_1]$, where P_1 is the premium for the put option written and P_2 is the premium for the put option held.

Scenario 3: (ST lies in between X_1 and X_2): Only the put option being held with a strike price of X_2 would be exercised while the put option being written with a strike price of X_1 would lapse and expire worthless. Total payoff is $(X_2 - ST)$ and profit would be: $[(X_2 - ST) - P_2 + P_1]$.

Butterfly Spread

A **butterfly spread** involves **holding calls (puts) with a low and high strike price and writing two calls (puts) with some intermediate strike price.**

It involves using three strike prices. It combines both holding and writing call (or put) options to create a position with limited risk and limited profit potential. Investors hold one lower strike price option (in-the-money) (X_1), hold another higher strike price option (out-of-the-money) (X_3) and write two middle strike options (at-the-money) (X_2).

The objective of a butterfly spread is to profit from a market that remains relatively stable or within a defined range. It's used when you anticipate minimal price movement in the underlying asset.

The ideal market outlook for a butterfly spread is one of low to moderate volatility. You expect the price of the underlying asset to stay near a specific strike price (the middle strike – X_2) without experiencing significant upward or downward movements.

There are now four scenarios:

Neutral Scenario: All the call options would expires worthless without being exercised. Total payoff would be zero and profit would be: $0 - C_1 - C_3 + 2C_2$, where C_1 is the premium for the call option held with strike price X_1 , C_3 is the premium for the call option held with strike price X_3 and C_2 is the premium for the two call options written at X_2 .

$ST < X_1$ Scenario: All the call options would expires worthless without being exercised. Total payoff would be zero and profit would be: $0 - C_1 - C_3 + 2C_2$, where C_1 is the premium for the call option held

with strike price X_1 , C_3 is the premium for the call option held with strike price X_3 and C_2 is the premium for the two call options written at X_2 .

ST lies between X_1 and X_2 Scenario: Only the call option held with the lowest strike price X_1 would be exercised, the other 3 would all lapse and expire worthless. Total Payoff would be $(ST - X_1)$ and profit would be: $(ST - X_1) - C_1 - C_3 + 2C_2$, where C_1 is the premium for the call option held with strike price X_1 , C_3 is the premium for the call option held with strike price X_3 and C_2 is the premium for the two call options written at X_2 .

ST lies between X_2 and X_3 Scenario: The call option held with the lowest strike price X_1 would be exercised and the two call options written with strike price X_2 would be exercised. The call option held with the highest strike price X_3 would lapse and expire worthless. Total payoff would be $(X_3 - ST)$ and profit would be: $(X_3 - ST) - C_1 - C_3 + 2C_2$, where C_1 is the premium for the call option held with strike price X_1 , C_3 is the premium for the call option held with strike price X_3 and C_2 is the premium for the two call options written at X_2 .

ST > X_3 Scenario: All four call options would be exercised. Total payoff would be zero and profit would be: $0 - C_1 - C_3 + 2C_2$, where C_1 is the premium for the call option held with strike price X_1 , C_3 is the premium for the call option held with strike price X_3 and C_2 is the premium for the two call options written at X_2 .